12 essays, 37,632 words. all of them are at paulgraham.com, free, as they should be. this is one reader's copy, printed at home to be read away from a screen, with a pencil.

made with on-paper. github.com/workinprogmess/on-paper

01

how to make wealth

paulgraham.com/wealth.html

if $3 million a year seems high to some people, it will seem low to others. three *million?* how do i get to be a billionaire, like bill gates?

so let's get bill gates out of the way right now. it's not a good idea to use famous rich people as examples, because the press only write about the very richest, and these tend to be outliers. bill gates is a smart, determined, and hardworking man, but you need more than that to make as much money as he has. you also need to be very lucky.

there is a large random factor in the success of any company. so the guys you end up reading about in the papers are the ones who are very smart, totally dedicated, *and* win the lottery. certainly bill is smart and dedicated, but microsoft also happens to have been the beneficiary of one of the most spectacular blunders in the history of business: the licensing deal for dos. no doubt bill did everything he could to steer ibm into making that blunder, and he has done an excellent job of exploiting it, but if there had been one person with a brain on ibm's side, microsoft's future would have been very different. microsoft at that stage had little leverage over ibm. they were effectively a component supplier. if ibm had required an exclusive license, as they should have, microsoft would still have signed the deal. it would still have meant a lot of money for them, and ibm could easily have gotten an operating system elsewhere.

instead ibm ended up using all its power in the market to give microsoft control of the pc standard. from that point, all microsoft had to do was execute. they never had to bet the company on a bold decision. all they had to do was play hardball with licensees and copy more innovative products reasonably promptly.

if ibm hadn't made this mistake, microsoft would still have been a successful company, but it could not have grown so big so fast. bill gates would be rich, but he'd be somewhere near the bottom of the forbes 400 with the other guys his age.

there are a lot of ways to get rich, and this essay is about only one of them. this essay is about how to make money by creating wealth and getting paid for it. there are plenty of other ways to get money, including chance, speculation, marriage, inheritance, theft, extortion, fraud, monopoly, graft, lobbying, counterfeiting, and prospecting. most of the greatest fortunes have probably involved several of these.

the advantage of creating wealth, as a way to get rich, is not just that it's more legitimate (many of the other methods are now illegal) but that it's more *straightforward.* you just have to do something people want.

money is not wealth

if you want to create wealth, it will help to understand what it is. wealth is not the same thing as money.[3] wealth is as old as human history. far older, in fact; ants have wealth. money is a comparatively recent invention.

wealth is the fundamental thing. wealth is stuff we want: food, clothes, houses, cars, gadgets, travel to interesting places, and so on. you can have wealth without having money. if you had a magic machine that could on command make you a car or cook you dinner or do your laundry, or do anything else you wanted, you wouldn't need money. whereas if you were in the middle of antarctica, where there is nothing to buy, it wouldn't matter how much money you had.

wealth is what you want, not money. but if wealth is the important thing, why does everyone talk about making money? it is a kind of shorthand: money is a way of moving wealth, and in practice they

think there was.[5]

kids know, without knowing they know, that they can create wealth. if you need to give someone a present and don't have any money, you make one. but kids are so bad at making things that they consider home-made presents to be a distinct, inferior, sort of thing to store-bought ones-- a mere expression of the proverbial thought that counts. and indeed, the lumpy ashtrays we made for our parents did not have much of a resale market.

craftsmen

the people most likely to grasp that wealth can be created are the ones who are good at making things, the craftsmen. their hand-made objects become store-bought ones. but with the rise of industrialization there are fewer and fewer craftsmen. one of the biggest remaining groups is computer programmers.

a programmer can sit down in front of a computer and *create wealth*. a good piece of software is, in itself, a valuable thing. there is no manufacturing to confuse the issue. those characters you type are a complete, finished product. if someone sat down and wrote a web browser that didn't suck (a fine idea, by the way), the world would be that much richer. [5b]

everyone in a company works together to create wealth, in the sense of making more things people want. many of the employees (e.g. the people in the mailroom or the personnel department) work at one remove from the actual making of stuff. not the programmers. they literally think the product, one line at a time. and so it's clearer to programmers that wealth is something that's made, rather than being distributed, like slices of a pie, by some imaginary daddy.

it's also obvious to programmers that there are huge variations in the rate at which wealth is created. at viaweb we had one programmer who was a sort of monster of productivity. i remember watching what he did one long day and estimating that he had added several hundred thousand dollars to the market value of the company. a great programmer, on a roll, could create a million dollars worth of wealth in a couple weeks. a mediocre programmer over the same period will generate zero or even negative wealth (e.g. by introducing bugs).

this is why so many of the best programmers are libertarians. in our world, you sink or swim, and there are no excuses. when those far removed from the creation of wealth-- undergraduates, reporters, politicians-- hear that the richest 5% of the people have half the total wealth, they tend to think *injustice!* an experienced programmer would be more likely to think *is that all?* the top 5% of programmers probably write 99% of the good software.

wealth can be created without being sold. scientists, till recently at least, effectively donated the wealth they created. we are all richer for knowing about penicillin, because we're less likely to die from infections. wealth is whatever people want, and not dying is certainly something we want. hackers often donate their work by writing open source software that anyone can use for free. i am much the richer for the operating system freebsd, which i'm running on the computer i'm using now, and so is yahoo, which runs it on all their servers.

what a job is

in industrialized countries, people belong to one institution or another at least until their twenties. after all those years you get used to the idea of belonging to a group of people who all get up in the morning, go to some set of buildings, and do things that they do not, ordinarily, enjoy doing. belonging to such a group becomes part of your identity: name, age, role, institution. if you have to introduce

companies are not set up to reward people who want to do this. you can't go to your boss and say, i'd like to start working ten times as hard, so will you please pay me ten times as much? for one thing, the official fiction is that you are already working as hard as you can. but a more serious problem is that the company has no way of measuring the value of your work.

salesmen are an exception. it's easy to measure how much revenue they generate, and they're usually paid a percentage of it. if a salesman wants to work harder, he can just start doing it, and he will automatically get paid proportionally more.

there is one other job besides sales where big companies can hire first-rate people: in the top management jobs. and for the same reason: their performance can be measured. the top managers are held responsible for the performance of the entire company. because an ordinary employee's performance can't usually be measured, he is not expected to do more than put in a solid effort. whereas top management, like salespeople, have to actually come up with the numbers. the ceo of a company that tanks cannot plead that he put in a solid effort. if the company does badly, he's done badly.

a company that could pay all its employees so straightforwardly would be enormously successful. many employees would work harder if they could get paid for it. more importantly, such a company would attract people who wanted to work especially hard. it would crush its competitors.

unfortunately, companies can't pay everyone like salesmen. salesmen work alone. most employees' work is tangled together. suppose a company makes some kind of consumer gadget. the engineers build a reliable gadget with all kinds of new features; the industrial designers design a beautiful case for it; and then the marketing people convince everyone that it's something they've got to have. how do you know how much of the gadget's sales are due to each group's efforts? or, for that matter, how much is due to the creators of past gadgets that gave the company a reputation for quality? there's no way to untangle all their contributions. even if you could read the minds of the consumers, you'd find these factors were all blurred together.

if you want to go faster, it's a problem to have your work tangled together with a large number of other people's. in a large group, your performance is not separately measurable-- and the rest of the group slows you down.

measurement and leverage

to get rich you need to get yourself in a situation with two things, measurement and leverage. you need to be in a position where your performance can be measured, or there is no way to get paid more by doing more. and you have to have leverage, in the sense that the decisions you make have a big effect.

measurement alone is not enough. an example of a job with measurement but not leverage is doing piecework in a sweatshop. your performance is measured and you get paid accordingly, but you have no scope for decisions. the only decision you get to make is how fast you work, and that can probably only increase your earnings by a factor of two or three.

an example of a job with both measurement and leverage would be lead actor in a movie. your performance can be measured in the gross of the movie. and you have leverage in the sense that your performance can make or break it.

ceos also have both measurement and leverage. they're measured, in that the performance of the company is their performance. and they have leverage in that their decisions set the whole company moving in one direction or another.

rather that small groups can be select. you don't want small in the sense of a village, but small in the sense of an all-star team.

the larger a group, the closer its average member will be to the average for the population as a whole. so all other things being equal, a very able person in a big company is probably getting a bad deal, because his performance is dragged down by the overall lower performance of the others. of course, all other things often are not equal: the able person may not care about money, or may prefer the stability of a large company. but a very able person who does care about money will ordinarily do better to go off and work with a small group of peers.

technology = leverage

startups offer anyone a way to be in a situation with measurement and leverage. they allow measurement because they're small, and they offer leverage because they make money by inventing new technology.

what is technology? it's *technique*. it's the way we all do things. and when you discover a new way to do things, its value is multiplied by all the people who use it. it is the proverbial fishing rod, rather than the fish. that's the difference between a startup and a restaurant or a barber shop. you fry eggs or cut hair one customer at a time. whereas if you solve a technical problem that a lot of people care about, you help everyone who uses your solution. that's leverage.

if you look at history, it seems that most people who got rich by creating wealth did it by developing new technology. you just can't fry eggs or cut hair fast enough. what made the florentines rich in 1200 was the discovery of new techniques for making the high-tech product of the time, fine woven cloth. what made the dutch rich in 1600 was the discovery of shipbuilding and navigation techniques that enabled them to dominate the seas of the far east.

fortunately there is a natural fit between smallness and solving hard problems. the leading edge of technology moves fast. technology that's valuable today could be worthless in a couple years. small companies are more at home in this world, because they don't have layers of bureaucracy to slow them down. also, technical advances tend to come from unorthodox approaches, and small companies are less constrained by convention.

big companies can develop technology. they just can't do it quickly. their size makes them slow and prevents them from rewarding employees for the extraordinary effort required. so in practice big companies only get to develop technology in fields where large capital requirements prevent startups from competing with them, like microprocessors, power plants, or passenger aircraft. and even in those fields they depend heavily on startups for components and ideas.

it's obvious that biotech or software startups exist to solve hard technical problems, but i think it will also be found to be true in businesses that don't seem to be about technology. mcdonald's, for example, grew big by designing a system, the mcdonald's franchise, that could then be reproduced at will all over the face of the earth. a mcdonald's franchise is controlled by rules so precise that it is practically a piece of software. write once, run everywhere. ditto for wal-mart. sam walton got rich not by being a retailer, but by designing a new kind of store.

use difficulty as a guide not just in selecting the overall aim of your company, but also at decision points along the way. at viaweb one of our rules of thumb was *run upstairs*. suppose you are a little, nimble guy being chased by a big, fat, bully. you open a door and find yourself in a staircase. do you go up or down? i say up. the bully can probably run downstairs as fast as you can. going upstairs his bulk will be more of a disadvantage. running upstairs is hard for you but even harder for him.

species, is that there are a lot of them, but this is little consolation to the individual mosquito.

startups, like mosquitos, tend to be an all-or-nothing proposition. and you don't generally know which of the two you're going to get till the last minute. viaweb came close to tanking several times. our trajectory was like a sine wave. fortunately we got bought at the top of the cycle, but it was damned close. while we were visiting yahoo in california to talk about selling the company to them, we had to borrow a conference room to reassure an investor who was about to back out of a new round of funding that we needed to stay alive.

the all-or-nothing aspect of startups was not something we wanted. viaweb's hackers were all extremely risk-averse. if there had been some way just to work super hard and get paid for it, without having a lottery mixed in, we would have been delighted. we would have much preferred a 100% chance of $1 million to a 20% chance of $10 million, even though theoretically the second is worth twice as much. unfortunately, there is not currently any space in the business world where you can get the first deal.

the closest you can get is by selling your startup in the early stages, giving up upside (and risk) for a smaller but guaranteed payoff. we had a chance to do this, and stupidly, as we then thought, let it slip by. after that we became comically eager to sell. for the next year or so, if anyone expressed the slightest curiosity about viaweb we would try to sell them the company. but there were no takers, so we had to keep going.

it would have been a bargain to buy us at an early stage, but companies doing acquisitions are not looking for bargains. a company big enough to acquire startups will be big enough to be fairly conservative, and within the company the people in charge of acquisitions will be among the more conservative, because they are likely to be business school types who joined the company late. they would rather overpay for a safe choice. so it is easier to sell an established startup, even at a large premium, than an early-stage one.

get users

i think it's a good idea to get bought, if you can. running a business is different from growing one. it is just as well to let a big company take over once you reach cruising altitude. it's also financially wiser, because selling allows you to diversify. what would you think of a financial advisor who put all his client's assets into one volatile stock?

how do you get bought? mostly by doing the same things you'd do if you didn't intend to sell the company. being profitable, for example. but getting bought is also an art in its own right, and one that we spent a lot of time trying to master.

potential buyers will always delay if they can. the hard part about getting bought is getting them to act. for most people, the most powerful motivator is not the hope of gain, but the fear of loss. for potential acquirers, the most powerful motivator is the prospect that one of their competitors will buy you. this, as we found, causes ceos to take red-eyes. the second biggest is the worry that, if they don't buy you now, you'll continue to grow rapidly and will cost more to acquire later, or even become a competitor.

in both cases, what it all comes down to is users. you'd think that a company about to buy you would do a lot of research and decide for themselves how valuable your technology was. not at all. what they go by is the number of users you have.

in effect, acquirers assume the customers know who has the best technology. and this is not as stupid as it sounds. users are the only real proof that you've created wealth. wealth is what people want, and

to a lesser extent britain under the labor governments of the 1960s and early 1970s. take away the incentive of wealth, and technical innovation grinds to a halt.

remember what a startup is, economically: a way of saying, i want to work faster. instead of accumulating money slowly by being paid a regular wage for fifty years, i want to get it over with as soon as possible. so governments that forbid you to accumulate wealth are in effect decreeing that you work slowly. they're willing to let you earn $3 million over fifty years, but they're not willing to let you work so hard that you can do it in two. they are like the corporate boss that you can't go to and say, i want to work ten times as hard, so please pay me ten times a much. except this is not a boss you can escape by starting your own company.

the problem with working slowly is not just that technical innovation happens slowly. it's that it tends not to happen at all. it's only when you're deliberately looking for hard problems, as a way to use speed to the greatest advantage, that you take on this kind of project. developing new technology is a pain in the ass. it is, as edison said, one percent inspiration and ninety-nine percent perspiration. without the incentive of wealth, no one wants to do it. engineers will work on sexy projects like fighter planes and moon rockets for ordinary salaries, but more mundane technologies like light bulbs or semiconductors have to be developed by entrepreneurs.

startups are not just something that happened in silicon valley in the last couple decades. since it became possible to get rich by creating wealth, everyone who has done it has used essentially the same recipe: measurement and leverage, where measurement comes from working with a small group, and leverage from developing new techniques. the recipe was the same in florence in 1200 as it is in santa clara today.

understanding this may help to answer an important question: why europe grew so powerful. was it something about the geography of europe? was it that europeans are somehow racially superior? was it their religion? the answer (or at least the proximate cause) may be that the europeans rode on the crest of a powerful new idea: allowing those who made a lot of money to keep it.

once you're allowed to do that, people who want to get rich can do it by generating wealth instead of stealing it. the resulting technological growth translates not only into wealth but into military power. the theory that led to the stealth plane was developed by a soviet mathematician. but because the soviet union didn't have a computer industry, it remained for them a theory; they didn't have hardware capable of executing the calculations fast enough to design an actual airplane.

in that respect the cold war teaches the same lesson as world war ii and, for that matter, most wars in recent history. don't let a ruling class of warriors and politicians squash the entrepreneurs. the same recipe that makes individuals rich makes countries powerful. let the nerds keep their lunch money, and you rule the world.

---

notes

1 one valuable thing you tend to get only in startups is *uninterruptability*. different kinds of work have different time quanta. someone proofreading a manuscript could probably be interrupted every fifteen minutes with little loss of productivity. but the time quantum for hacking is very long: it might take an hour just to load a problem into your head. so the cost of having someone from personnel call you about a form you forgot to fill out can be huge.

hing (the automobile, the television, the computer, the jet engine, the laser), it's because other companies made all the money.

9 this is a good plan for life in general. if you have two choices, choose the harder. if you're trying to decide whether to go out running or sit home and watch tv, go running. probably the reason this trick works so well is that when you have two choices and one is harder, the only reason you're even considering the other is laziness. you know in the back of your mind what's the right thing to do, and this trick merely forces you to acknowledge it.

10 it is probably no accident that the middle class first appeared in northern italy and the low countries, where there were no strong central governments. these two regions were the richest of their time and became the twin centers from which renaissance civilization radiated. if they no longer play that role, it is because other places, like the united states, have been truer to the principles they discovered.

11 it may indeed be a sufficient condition. but if so, why didn't the industrial revolution happen earlier? two possible (and not incompatible) answers: (a) it did. the industrial revolution was one in a series. (b) because in medieval towns, monopolies and guild regulations initially slowed the development of new means of production.

comment on this essay.

02

how people get rich now

april 2021
paulgraham.com/richnow.html

natural resources, or doing real estate deals. is that really better than a world in which the richest people get rich by starting tech companies?

why are people starting so many more new companies than they used to, and why are they getting so rich from it? the answer to the first question, curiously enough, is that it's misphrased. we shouldn't be asking why people are starting companies, but why they're starting companies *again*. [3]

in 1892, the *new york herald tribune* compiled a list of all the millionaires in america. they found 4047 of them. how many had inherited their wealth then? only about 20%, which is less than the proportion of heirs today. and when you investigate the sources of the new fortunes, 1892 looks even more like today. hugh rockoff found that "many of the richest ... gained their initial edge from the new technology of mass production." [4]

so it's not 2020 that's the anomaly here, but 1982. the real question is why so few people had gotten rich from starting companies in 1982. and the answer is that even as the *herald tribune*'s list was being compiled, a wave of consolidation was sweeping through the american economy. in the late 19th and early 20th centuries, financiers like j. p. morgan combined thousands of smaller companies into a few hundred giant ones with commanding economies of scale. by the end of world war ii, as michael lind writes, "the major sectors of the economy were either organized as government-backed cartels or dominated by a few oligopolistic corporations." [5]

in 1960, most of the people who start startups today would have gone to work for one of them. you could get rich from starting your own company in 1890 and in 2020, but in 1960 it was not really a viable option. you couldn't break through the oligopolies to get at the markets. so the prestigious route in 1960 was not to start your own company, but to work your way up the corporate ladder at an existing one. [6]

making everyone a corporate employee decreased economic inequality (and every other kind of variation), but if your model of normal is the mid 20th century, you have a very misleading model in that respect. j. p. morgan's economy turned out to be just a phase, and starting in the 1970s, it began to break up.

why did it break up? partly senescence. the big companies that seemed models of scale and efficiency in 1930 had by 1970 become slack and bloated. by 1970 the rigid structure of the economy was full of cosy nests that various groups had built to insulate themselves from market forces. during the carter administration the federal government realized something was amiss and began, in a process they called "deregulation," to roll back the policies that propped up the oligopolies.

but it wasn't just decay from within that broke up j. p. morgan's economy. there was also pressure from without, in the form of new technology, and particularly microelectronics. the best way to envision what happened is to imagine a pond with a crust of ice on top. initially the only way from the bottom to the surface is around the edges. but as the ice crust weakens, you start to be able to punch right through the middle.

the edges of the pond were pure tech: companies that actually described themselves as being in the electronics or software business. when you used the word "startup" in 1990, that was what you meant. but now startups are punching right through the middle of the ice crust and displacing incumbents like retailers and tv networks and car companies. [7]

but though the breakup of j. p. morgan's economy created a new world in the technological sense, it was a reversion to the norm in the social sense. if you only look back as far as the mid 20th century, it seems like people getting rich by starting their own companies is a recent phenomenon. but if you look

3 when i say people are starting more companies, i mean the type of company meant to grow very big. there has actually been a decrease in the last couple decades in the overall number of new companies. but the vast majority of companies are small retail and service businesses. so what the statistics about the decreasing number of new businesses mean is that people are starting fewer shoe stores and barber shops.

people sometimes get confused when they see a graph labelled "startups" that's going down, because there are two senses of the word "startup": (1) the founding of a company, and (2) a particular type of company designed to grow big fast. the statistics mean startup in sense (1), not sense (2).

4 rockoff, hugh. "great fortunes of the gilded age." nber working paper 14555, 2008.

5 lind, michael. *land of promise.* harpercollins, 2012.

it's also likely that the high tax rates in the mid 20th century deterred people from starting their own companies. starting one's own company is risky, and when risk isn't rewarded, people opt for safety instead.

but it wasn't simply cause and effect. the oligopolies and high tax rates of the mid 20th century were all of a piece. lower taxes are not just a cause of entrepreneurship, but an effect as well: the people getting rich in the mid 20th century from real estate and oil exploration lobbied for and got huge tax loopholes that made their effective tax rate much lower, and presumably if it had been more common to grow big companies by building new technology, the people doing that would have lobbied for their own loopholes as well.

6 that's why the people who did get rich in the mid 20th century so often got rich from oil exploration or real estate. those were the two big areas of the economy that weren't susceptible to consolidation.

7 the pure tech companies used to be called "high technology" startups. but now that startups can punch through the middle of the ice crust, we don't need a separate name for the edges, and the term "high-tech" has a decidedly retro sound.

8 higher valuations mean you either sell less stock to get a given amount of money, or get more money for a given amount of stock. the typical startup does some of each. obviously you end up richer if you keep more stock, but you should also end up richer if you raise more money, because (a) it should make the company more successful, and (b) you should be able to last longer before the next round, or not even need one. notice all those shoulds though. in practice a lot of money slips through them.

it might seem that the huge rounds raised by startups nowadays contradict the claim that it has become cheaper to start one. but there's no contradiction here; the startups that raise the most are the ones doing it by choice, in order to grow faster, not the ones doing it because they need the money to survive. there's nothing like not needing money to make people offer it to you.

you would think, after having been on the side of labor in its fight with capital for almost two centuries, that the far left would be happy that labor has finally prevailed. but none of them seem to be. you can almost hear them saying "no, no, not *that* way."

9 ibm was created in 1911 by merging three companies, the most important of which was herman hollerith's tabulating machine company, founded in 1896. in 1941 its revenues were $60 million.

hewlett-packard's revenues in 1964 were $125 million.

microsoft's revenues in 1988 were $590 million.

thanks to trevor blackwell, jessica livingston, bob lesko, robert morris, russ roberts, and alex tabarrok for reading drafts of this, and to jon erlichman for growth data.

03

billionaires build

december 2020
paulgraham.com/ace.html

by growing a small market. this idea is important enough that it's worth coining a phrase for, so let's call one of these small but growable markets a "larval market."

the perfect example of a larval market might be apple's market when they were founded in 1976. in 1976, not many people wanted their own computer. but more and more started to want one, till now every 10 year old on the planet wants a computer (but calls it a "phone").

the ideal combination is the group of founders who are "living in the future" in the sense of being at the leading edge of some kind of change, and who are building something they themselves want. most super-successful startups are of this type. steve wozniak wanted a computer. mark zuckerberg wanted to engage online with his college friends. larry and sergey wanted to find things on the web. all these founders were building things they and their peers wanted, and the fact that they were at the leading edge of change meant that more people would want these things in the future.

but although the ideal larval market is oneself and one's peers, that's not the only kind. a larval market might also be regional, for example. you build something to serve one location, and then expand to others.

the crucial feature of the initial market is that it exist. that may seem like an obvious point, but the lack of it is the biggest flaw in most startup ideas. there have to be some people who want what you're building right now, and want it so urgently that they're willing to use it, bugs and all, even though you're a small company they've never heard of. there don't have to be many, but there have to be some. as long as you have some users, there are straightforward ways to get more: build new features they want, seek out more people like them, get them to refer you to their friends, and so on. but these techniques all require some initial seed group of users.

so this is one thing the yc partners will almost certainly dig into during your interview. who are your first users going to be, and how do you know they want this? if i had to decide whether to fund startups based on a single question, it would be "how do you know people want this?"

the most convincing answer is "because we and our friends want it." it's even better when this is followed by the news that you've already built a prototype, and even though it's very crude, your friends are using it, and it's spreading by word of mouth. if you can say that and you're not lying, the partners will switch from default no to default yes. meaning you're in unless there's some other disqualifying flaw.

that is a hard standard to meet, though. airbnb didn't meet it. they had the first part. they had made something they themselves wanted. but it wasn't spreading. so don't feel bad if you don't hit this gold standard of convincingness. if airbnb didn't hit it, it must be too high.

in practice, the yc partners will be satisfied if they feel that you have a deep understanding of your users' needs. and the airbnbs did have that. they were able to tell us all about what motivated hosts and guests. they knew from first-hand experience, because they'd been the first hosts. we couldn't ask them a question they didn't know the answer to. we ourselves were not very excited about the idea as users, but we knew this didn't prove anything, because there were lots of successful startups we hadn't been excited about as users. we were able to say to ourselves "they seem to know what they're talking about. maybe they're onto something. it's not growing yet, but maybe they can figure out how to make it grow during yc." which they did, about three weeks into the batch.

the best thing you can do in a yc interview is to teach the partners about your users. so if you want to prepare for your interview, one of the best ways to do it is to go talk to your users and find out exactly what they're thinking. which is what you should be doing anyway.

your idea seem perfect causes you to come off as glib or clueless, you've sacrificed something you needed for something you didn't.

if the partners are sufficiently convinced that there's a path to a big market, the next question is whether you'll be able to find it. that in turn depends on three things: the general qualities of the founders, their specific expertise in this domain, and the relationship between them. how determined are the founders? are they good at building things? are they resilient enough to keep going when things go wrong? how strong is their friendship?

though the airbnbs only did ok in the idea department, they did spectacularly well in this department. the story of how they'd funded themselves by making obama- and mccain-themed breakfast cereal was the single most important factor in our decision to fund them. they didn't realize it at the time, but what seemed to them an irrelevant story was in fact fabulously good evidence of their qualities as founders. it showed they were resourceful and determined, and could work together.

it wasn't just the cereal story that showed that, though. the whole interview showed that they cared. they weren't doing this just for the money, or because startups were cool. the reason they were working so hard on this company was because it was their project. they had discovered an interesting new idea, and they just couldn't let it go.

mundane as it sounds, that's the most powerful motivator of all, not just in startups, but in most ambitious undertakings: to be genuinely interested in what you're building. this is what really drives billionaires, or at least the ones who become billionaires from starting companies. the company is their project.

one thing few people realize about billionaires is that all of them could have stopped sooner. they could have gotten acquired, or found someone else to run the company. many founders do. the ones who become really rich are the ones who keep working. and what makes them keep working is not just money. what keeps them working is the same thing that keeps anyone else working when they could stop if they wanted to: that there's nothing else they'd rather do.

that, not exploiting people, is the defining quality of people who become billionaires from starting companies. so that's what yc looks for in founders: authenticity. people's motives for starting startups are usually mixed. they're usually doing it from some combination of the desire to make money, the desire to seem cool, genuine interest in the problem, and unwillingness to work for someone else. the last two are more powerful motivators than the first two. it's ok for founders to want to make money or to seem cool. most do. but if the founders seem like they're doing it *just* to make money or *just* to seem cool, they're not likely to succeed on a big scale. the founders who are doing it for the money will take the first sufficiently large acquisition offer, and the ones who are doing it to seem cool will rapidly discover that there are much less painful ways of seeming cool. [3]

y combinator certainly sees founders whose m.o. is to exploit people. yc is a magnet for them, because they want the yc brand. but when the yc partners detect someone like that, they reject them. if bad people made good founders, the yc partners would face a moral dilemma. fortunately they don't, because bad people make bad founders. this exploitative type of founder is not going to succeed on a large scale, and in fact probably won't even succeed on a small one, because they're always going to be taking shortcuts. they see yc itself as a shortcut.

their exploitation usually begins with their own cofounders, which is disastrous, since the cofounders' relationship is the foundation of the company. then it moves on to the users, which is also disastrous, because the sort of early adopters a successful startup wants as its initial users are the hardest to fool.

for some people startups are the most efficient way to do it. i'm just saying that these are not the startups that get really big.

4 not these days, anyway. there were some big ones during the internet bubble, and indeed some big ipos.

thanks to trevor blackwell, jessica livingston, robert morris, geoff ralston, and harj taggar for reading drafts of this.

04

ramen profitable

paulgraham.com/ramenprofitable.html

one. when people start to pay you significant amounts, the company starts to feel real. and your own living expenses are the milestone you feel most, because at that point the future flips state. now survival is the default, instead of dying.

a morale boost on that scale is very valuable in a startup, because the moral weight of running a startup is what makes it hard. startups are still very rare. why don't more people do it? the financial risk? plenty of 25 year olds save nothing anyway. the long hours? plenty of people work just as long hours in regular jobs. what keeps people from starting startups is the fear of having so much responsibility. and this is not an irrational fear: it really is hard to bear. anything that takes some of that weight off you will greatly increase your chances of surviving.

a startup that reaches ramen profitability may be more likely to succeed than not. which is pretty exciting, considering the bimodal distribution of outcomes in startups: you either fail or make a lot of money.

the fourth advantage of ramen profitability is the least obvious but may be the most important. if you don't need to raise money, you don't have to interrupt working on the company to do it.

raising money is terribly distracting. you're lucky if your productivity is a third of what it was before. and it can last for months.

i didn't understand (or rather, remember) precisely why raising money was so distracting till earlier this year. i'd noticed that startups we funded would usually grind to a halt when they switched to raising money, but i didn't remember exactly why till yc raised money itself. we had a comparatively easy time of it; the first people i asked said yes; but it took months to work out the details, and during that time i got hardly any real work done. why? because i thought about it all the time.

at any given time there tends to be one problem that's the most urgent for a startup. this is what you think about as you fall asleep at night and when you take a shower in the morning. and when you start raising money, that becomes the problem you think about. you only take one shower in the morning, and if you're thinking about investors during it, then you're not thinking about the product.

whereas if you can choose when you raise money, you can pick a time when you're not in the middle of something else, and you can probably also insist that the round close fast. you may even be able to avoid having the round occupy your thoughts, if you don't care whether it closes.

* * *

ramen profitable means no more than the definition implies. it does not, for example, imply that you're "bootstrapping" the startup—that you're never going to take money from investors. empirically that doesn't seem to work very well. few startups succeed without taking investment. maybe as startups get cheaper it will become more common. on the other hand, the money is there, waiting to be invested. if startups need it less, they'll be able to get it on better terms, which will make them more inclined to take it. that will tend to produce an equilibrium. [2]

another thing ramen profitability doesn't imply is joe kraus's idea that you should put your business model in beta when you put your product in beta. he believes you should get people to pay you from the beginning. i think that's too constraining. facebook didn't, and they've done better than most startups. making money right away was not only unnecessary for them, but probably would have been harmful. i do think joe's rule could be useful for many startups, though. when founders seem unfocused, i sometimes suggest they try to get customers to pay them for something, in the hope that this constraint will prod them into action.

05

default alive or default dead?

october 2015
paulgraham.com/aord.html

if you're paying attention, you'll be asking at this point not just how to avoid the fatal pinch, but how to avoid being default dead. that one is easy: don't hire too fast. hiring too fast is by far the biggest killer of startups that raise money. [2]

founders tell themselves they need to hire in order to grow. but most err on the side of overestimating this need rather than underestimating it. why? partly because there's so much work to do. naive founders think that if they can just hire enough people, it will all get done. partly because successful startups have lots of employees, so it seems like that's what one does in order to be successful. in fact the large staffs of successful startups are probably more the effect of growth than the cause. and partly because when founders have slow growth they don't want to face what is usually the real reason: the product is not appealing enough.

plus founders who've just raised money are often encouraged to overhire by the vcs who funded them. kill-or-cure strategies are optimal for vcs because they're protected by the portfolio effect. vcs want to blow you up, in one sense of the phrase or the other. but as a founder your incentives are different. you want above all to survive. [3]

here's a common way startups die. they make something moderately appealing and have decent initial growth. they raise their first round fairly easily, because the founders seem smart and the idea sounds plausible. but because the product is only moderately appealing, growth is ok but not great. the founders convince themselves that hiring a bunch of people is the way to boost growth. their investors agree. but (because the product is only moderately appealing) the growth never comes. now they're rapidly running out of runway. they hope further investment will save them. but because they have high expenses and slow growth, they're now unappealing to investors. they're unable to raise more, and the company dies.

what the company should have done is address the fundamental problem: that the product is only moderately appealing. hiring people is rarely the way to fix that. more often than not it makes it harder. at this early stage, the product needs to evolve more than to be "built out," and that's usually easier with fewer people. [4]

asking whether you're default alive or default dead may save you from this. maybe the alarm bells it sets off will counteract the forces that push you to overhire. instead you'll be compelled to seek growth in other ways. for example, by doing things that don't scale, or by redesigning the product in the way only founders can. and for many if not most startups, these paths to growth will be the ones that actually work.

airbnb waited 4 months after raising money at the end of y combinator before they hired their first employee. in the meantime the founders were terribly overworked. but they were overworked evolving airbnb into the astonishingly successful organism it is now.

---

notes

1 steep usage growth will also interest investors. revenue will ultimately be a constant multiple of usage, so x% usage growth predicts x% revenue growth. but in practice investors discount merely predicted revenue, so if you're measuring usage you need a higher growth rate to impress investors.

2 startups that don't raise money are saved from hiring too fast because they can't afford to. but that doesn't mean you should avoid raising money in order to avoid this problem, any more than that total abstinence is

06

the fatal pinch

december 2014
paulgraham.com/pinch.html

getting people to take less salary is a weak solution that will only work when the problem isn't too bad. if your current trajectory won't quite get you to profitability but you can get over the threshold by cutting salaries a little, you might be able to make the case to everyone for doing it. otherwise you're probably just postponing the problem, and that will be obvious to the people whose salaries you're proposing to cut. [4]

which leaves two options, firing good people and making more money. while trying to balance them, keep in mind the eventual goal: to be a successful product company in the sense of having a single thing lots of people use.

you should lean more toward firing people if the source of your trouble is overhiring. if you went out and hired 15 people before you even knew what you were building, you've created a broken company. you need to figure out what you're building, and it will probably be easier to do that with a handful of people than 15. plus those 15 people might not even be the ones you need for whatever you end up building. so the solution may be to shrink and then figure out what direction to grow in. after all, you're not doing those 15 people any favors if you fly the company into ground with them aboard. they'll all lose their jobs eventually, along with all the time they expended on this doomed company.

whereas if you only have a handful of people, it may be better to focus on trying to make more money. it may seem facile to suggest a startup make more money, as if that could be done for the asking. usually a startup is already trying as hard as it can to sell whatever it sells. what i'm suggesting here is not so much to try harder to make money but to try to make money in a different way. for example, if you have only one person selling while the rest are writing code, consider having everyone work on selling. what good will more code do you when you're out of business? if you have to write code to close a certain deal, go ahead; that follows from everyone working on selling. but only work on whatever will get you the most revenue the soonest.

another way to make money differently is to sell different things, and in particular to do more consultingish work. i say consultingish because there is a long slippery slope from making products to pure consulting, and you don't have to go far down it before you start to offer something really attractive to customers. although your product may not be very appealing yet, if you're a startup your programmers will often be way better than the ones your customers have. or you may have expertise in some new field they don't understand. so if you change your sales conversations just a little from "do you want to buy our product?" to "what do you need that you'd pay a lot for?" you may find it's suddenly a lot easier to extract money from customers.

be ruthlessly mercenary when you start doing this, though. you're trying to save your company from death here, so make customers pay a lot, quickly. and to the extent you can, try to avoid the worst pitfalls of consulting. the ideal thing might be if you built a precisely defined derivative version of your product for the customer, and it was otherwise a straight product sale. you keep the ip and no billing by the hour.

in the best case, this consultingish work may not be just something you do to survive, but may turn out to be the thing-that-doesn't-scale that defines your company. don't expect it to be, but as you dive into individual users' needs, keep your eyes open for narrow openings that have wide vistas beyond.

there is usually so much demand for custom work that unless you're really incompetent there has to be some point down the slope of consulting at which you can survive. but i didn't use the term slippery slope by accident; customers' insatiable demand for custom work will always be pushing you toward

07

the equity equation

july 2007
paulgraham.com/equity.html

i think to translate salary and overhead into stock you should multiply the annual rate by about 1.5. most startups grow fast or die; if you die you don't have to pay the guy, and if you grow fast you'll be paying next year's salary out of next year's valuation, which should be 3x this year's. if your valuation grows 3x a year, the total cost in stock of a new hire's salary and overhead is 1.5 years' cost at the present valuation.[2]

how much of an additional margin should the company need as the "activation energy" for the deal? since this is in effect the company's profit on a hire, the market will determine that: if you're a hot opportunity, you can charge more.

let's run through an example. suppose the company wants to make a "profit" of 50% on the new hire mentioned above. so subtract a third from 16.7% and we have 11.1% as his "retail" price. suppose further that he's going to cost $60k a year in salary and overhead, x 1.5 = $90k total. if the company's valuation is $2 million, $90k is 4.5%. 11.1% - 4.5% = an offer of 6.6%.

incidentally, notice how important it is for early employees to take little salary. it comes right out of stock that could otherwise be given to them.

obviously there is a great deal of play in these numbers. i'm not claiming that stock grants can now be reduced to a formula. ultimately you always have to guess. but at least know what you're guessing. if you choose a number based on your gut feel, or a table of typical grant sizes supplied by a vc firm, understand what those are estimates of.

and more generally, when you make any decision involving equity, run it through 1/(1 - n) to see if it makes sense. you should always feel richer after trading equity. if the trade didn't increase the value of your remaining shares enough to put you net ahead, you wouldn't have (or shouldn't have) done it.

---

notes

1 this is why we can't believe anyone would think y combinator was a bad deal. does anyone really think we're so useless that in three months we can't improve a startup's prospects by 7.5%?

2 the obvious choice for your present valuation is the post-money valuation of your last funding round. this probably undervalues the company, though, because (a) unless your last round just happened, the company is presumably worth more, and (b) the valuation of an early funding round usually reflects some other contribution by the investors.

thanks to sam altman, trevor blackwell, paul buchheit, hutch fishman, david hornik, paul kedrosky, jessica livingston, gary sabot, and joshua schachter for reading drafts of this.

08

investor herd dynamics

paulgraham.com/herd.html

unless you're a wizard at negotiation (and if you're not sure, you're not) be very careful about exaggerating this to push a good investor to decide. founders try this sort of thing all the time, and investors are very sensitive to it. if anything oversensitive. but you're safe so long as you're telling the truth. if you're getting far along with investor b, but you'd rather raise money from investor a, you can tell investor a that this is happening. there's no manipulation in that. you're genuinely in a bind, because you really would rather raise money from a, but you can't safely reject an offer from b when it's still uncertain what a will decide.

do not, however, tell a who b is. vcs will sometimes ask which other vcs you're talking to, but you should never tell them. angels you can sometimes tell about other angels, because angels cooperate more with one another. but if vcs ask, just point out that they wouldn't want you telling other firms about your conversations, and you feel obliged to do the same for any firm you talk to. if they push you, point out that you're inexperienced at fundraising — which is always a safe card to play — and you feel you have to be extra cautious. [3]

while few startups will experience a stampede of interest, almost all will at least initially experience the other side of this phenomenon, where the herd remains clumped together at a distance. the fact that investors are so much influenced by other investors' opinions means you always start out in something of a hole. so don't be demoralized by how hard it is to get the first commitment, because much of the difficulty comes from this external force. the second will be easier.

---

notes

1 an accountant might say that a company that has raised a million dollars is no richer if it's convertible debt, but in practice money raised as convertible debt is little different from money raised in an equity round.

2 founders are often surprised by this, but investors can get very emotional. or rather indignant; that's the main emotion i've observed; but it is very common, to the point where it sometimes causes investors to act against their own interests. i know of one investor who invested in a startup at a $15 million valuation cap. earlier he'd had an opportunity to invest at a $5 million cap, but he refused because a friend who invested earlier had been able to invest at a $3 million cap.

3 if an investor pushes you hard to tell them about your conversations with other investors, is this someone you want as an investor?

thanks to paul buchheit, jessica livingston, geoff ralston, and garry tan for reading drafts of this.

09

high resolution fundraising

paulgraham.com/hiresfund.html

the most interesting question here may be what high res fundraising will do to the world of investors. bolder investors will now get rewarded with lower prices. but more important, in a hits-driven business, is that they'll be able to get into the deals they want. whereas the "who else is investing?" type of investors will not only pay higher prices, but may not be able to get into the best deals at all.

thanks to immad akhund, sam altman, john bautista, pete koomen, jessica livingston, dan siroker, harj taggar, and fred wilson for reading drafts of this.

10

how to convince investors

paulgraham.com/convince.html

something you're not, you'll just end up in an uncanny valley. you'll depart from sincere, but never arrive at convincing.

truth

the way to seem most formidable as an inexperienced founder is to stick to the truth. how formidable you seem isn't a constant. it varies depending on what you're saying. most people can seem confident when they're saying "one plus one is two," because they know it's true. the most diffident person would be puzzled and even slightly contemptuous if they told a vc "one plus one is two" and the vc reacted with skepticism. the magic ability of people who are good at seeming formidable is that they can do this with the sentence "we're going to make a billion dollars a year." but you can do the same, if not with that sentence with some fairly impressive ones, so long as you convince yourself first.

that's the secret. convince yourself that your startup is worth investing in, and then when you explain this to investors they'll believe you. and by convince yourself, i don't mean play mind games with yourself to boost your confidence. i mean truly evaluate whether your startup is worth investing in. if it isn't, don't try to raise money. [5] but if it is, you'll be telling the truth when you tell investors it's worth investing in, and they'll sense that. you don't have to be a smooth presenter if you understand something well and tell the truth about it.

to evaluate whether your startup is worth investing in, you have to be a domain expert. if you're not a domain expert, you can be as convinced as you like about your idea, and it will seem to investors no more than an instance of the dunning-kruger effect. which in fact it will usually be. and investors can tell fairly quickly whether you're a domain expert by how well you answer their questions. know everything about your market. [6]

why do founders persist in trying to convince investors of things they're not convinced of themselves? partly because we've all been trained to.

when my friends robert morris and trevor blackwell were in grad school, one of their fellow students was on the receiving end of a question from their faculty advisor that we still quote today. when the unfortunate fellow got to his last slide, the professor burst out:

> which one of these conclusions do you actually believe?

one of the artifacts of the way schools are organized is that we all get trained to talk even when we have nothing to say. if you have a ten page paper due, then ten pages you must write, even if you only have one page of ideas. even if you have no ideas. you have to produce something. and all too many startups go into fundraising in the same spirit. when they think it's time to raise money, they try gamely to make the best case they can for their startup. most never think of pausing beforehand to ask whether what they're saying is actually convincing, because they've all been trained to treat the need to present as a given — as an area of fixed size, over which however much truth they have must needs be spread, however thinly.

the time to raise money is not when you need it, or when you reach some artificial deadline like a demo day. it's when you can convince investors, and not before. [7]

and unless you're a good con artist, you'll never convince investors if you're not convinced yourself. they're far better at detecting bullshit than you are at producing it, even if you're producing it unknowingly. if you try to convince investors before you've convinced yourself, you'll be wasting both your time.

if you can make as good a case as microsoft could have, will you convince investors? not always. a lot of vcs would have rejected microsoft. [9] certainly some rejected google. and getting rejected will put you in a slightly awkward position, because as you'll see when you start fundraising, the most common question you'll get from investors will be "who else is investing?" what do you say if you've been fundraising for a while and no one has committed yet? [10]

the people who are really good at acting formidable often solve this problem by giving investors the impression that while no investors have committed yet, several are about to. this is arguably a permissible tactic. it's slightly dickish of investors to care more about who else is investing than any other aspect of your startup, and misleading them about how far along you are with other investors seems the complementary countermove. it's arguably an instance of scamming a scammer. but i don't recommend this approach to most founders, because most founders wouldn't be able to carry it off. this is the single most common lie told to investors, and you have to be really good at lying to tell members of some profession the most common lie they're told.

if you're not a master of negotiation (and perhaps even if you are) the best solution is to tackle the problem head-on, and to explain why investors have turned you down and why they're mistaken. if you know you're on the right track, then you also know why investors were wrong to reject you. experienced investors are well aware that the best ideas are also the scariest. they all know about the vcs who rejected google. if instead of seeming evasive and ashamed about having been turned down (and thereby implicitly agreeing with the verdict) you talk candidly about what scared investors about you, you'll seem more confident, which they like, and you'll probably also do a better job of presenting that aspect of your startup. at the very least, that worry will now be out in the open instead of being a gotcha left to be discovered by the investors you're currently talking to, who will be proud of and thus attached to their discovery. [11]

this strategy will work best with the best investors, who are both hard to bluff and who already believe most other investors are conventional-minded drones doomed always to miss the big outliers. raising money is not like applying to college, where you can assume that if you can get into mit, you can also get into foobar state. because the best investors are much smarter than the rest, and the best startup ideas look initially like bad ideas, it's not uncommon for a startup to be rejected by all the vcs except the best ones. that's what happened to dropbox. y combinator started in boston, and for the first 3 years we ran alternating batches in boston and silicon valley. because boston investors were so few and so timid, we used to ship boston batches out for a second demo day in silicon valley. dropbox was part of a boston batch, which means all those boston investors got the first look at dropbox, and none of them closed the deal. yet another backup and syncing thing, they all thought. a couple weeks later, dropbox raised a series a round from sequoia. [12]

different

not understanding that investors view investments as bets combines with the ten page paper mentality to prevent founders from even considering the possibility of being certain of what they're saying. they think they're trying to convince investors of something very uncertain — that their startup will be huge — and convincing anyone of something like that must obviously entail some wild feat of salesmanship. but in fact when you raise money you're trying to convince investors of something so much less speculative — whether the company has all the elements of a good bet — that you can approach the problem in a qualitatively different way. you can convince yourself, then convince them.

and when you convince them, use the same matter-of-fact language you used to convince yourself. you wouldn't use vague, grandiose marketing-speak among yourselves. don't use it with investors either. it

raise money, it's not enough to be promising. you have to be delivering results.

so although it works well to show growth graphs at either stage, investors treat them differently. at three months, a growth graph is mostly evidence that the founders are effective. at two years, it has to be evidence of a promising market and a company tuned to exploit it.

9 by this i mean that if the present day equivalent of the 3 month old microsoft presented at a demo day, there would be investors who turned them down. microsoft itself didn't raise outside money, and indeed the venture business barely existed when they got started in 1975.

10 the best investors rarely care who else is investing, but mediocre investors almost all do. so you can use this question as a test of investor quality.

11 to use this technique, you'll have to find out why investors who rejected you did so, or at least what they claim was the reason. that may require asking, because investors don't always volunteer a lot of detail. make it clear when you ask that you're not trying to dispute their decision — just that if there is some weakness in your plans, you need to know about it. you won't always get a real reason out of them, but you should at least try.

12 dropbox wasn't rejected by all the east coast vcs. there was one firm that wanted to invest but tried to lowball them.

13 alfred lin points out that it's doubly important for the explanation of a startup to be clear and concise, because it has to convince at one remove: it has to work not just on the partner you talk to, but when that partner re-tells it to colleagues.

we consciously optimize for this at yc. when we work with founders create a demo day pitch, the last step is to imagine how an investor would sell it to colleagues.

thanks to marc andreessen, sam altman, patrick collison, ron conway, chris dixon, alfred lin, ben horowitz, steve huffman, jessica livingston, greg mcadoo, andrew mason, geoff ralston, yuri sagalov, emmett shear, rajat suri, garry tan, albert wenger, fred wilson, and qasar younis for reading drafts of this.

11

how to raise money

paulgraham.com/fr.html

such a high proportion of successful startups raise money that it might seem fundraising is one of the defining qualities of a startup. actually it isn't. rapid growth is what makes a company a startup. most companies in a position to grow rapidly find that (a) taking outside money helps them grow faster, and (b) their growth potential makes it easy to attract such money. it's so common for both (a) and (b) to be true of a successful startup that practically all do raise outside money. but there may be cases where a startup either wouldn't want to grow faster, or outside money wouldn't help them to, and if you're one of them, don't raise money.

the other time not to raise money is when you won't be able to. if you try to raise money before you can convince investors, you'll not only waste your time, but also burn your reputation with those investors.

be in fundraising mode or not.

one of the things that surprises founders most about fundraising is how distracting it is. when you start fundraising, everything else grinds to a halt. the problem is not the time fundraising consumes but that it becomes the top idea in your mind. a startup can't endure that level of distraction for long. an early stage startup grows mostly because the founders make it grow, and if the founders look away, growth usually drops sharply.

because fundraising is so distracting, a startup should either be in fundraising mode or not. and when you do decide to raise money, you should focus your whole attention on it so you can get it done quickly and get back to work. [2]

you can take money from investors when you're not in fundraising mode. you just can't expend any attention on it. there are two things that take attention: convincing investors, and negotiating with them. so when you're not in fundraising mode, you should take money from investors only if they require no convincing, and are willing to invest on terms you'll take without negotiation. for example, if a reputable investor is willing to invest on a convertible note, using standard paperwork, that is either uncapped or capped at a good valuation, you can take that without having to think. [3] the terms will be whatever they turn out to be in your next equity round. and "no convincing" means just that: zero time spent meeting with investors or preparing materials for them. if an investor says they're ready to invest, but they need you to come in for one meeting to meet some of the partners, tell them no, if you're not in fundraising mode, because that's fundraising. [4] tell them politely; tell them you're focusing on the company right now, and that you'll get back to them when you're fundraising; but do not get sucked down the slippery slope.

investors will try to lure you into fundraising when you're not. it's great for them if they can, because they can thereby get a shot at you before everyone else. they'll send you emails saying they want to meet to learn more about you. if you get cold-emailed by an associate at a vc firm, you shouldn't meet even if you are in fundraising mode. deals don't happen that way. [5] but even if you get an email from a partner you should try to delay meeting till you're in fundraising mode. they may say they just want to meet and chat, but investors never just want to meet and chat. what if they like you? what if they start to talk about giving you money? will you be able to resist having that conversation? unless you're experienced enough at fundraising to have a casual conversation with investors that stays casual, it's safer to tell them that you'd be happy to later, when you're fundraising, but that right now you need to focus on the company. [6]

companies that are successful at raising money in phase 2 sometimes tack on a few investors after leaving fundraising mode. this is fine; if fundraising went well, you'll be able to do it without spending

expected value = how likely an investor is to say yes, multiplied by how good it would be if they did. so for example, an eminent investor who would invest a lot, but will be hard to convince, might have the same expected value as an obscure angel who won't invest much, but will be easy to convince. whereas an obscure angel who will only invest a small amount, and yet needs to meet multiple times before making up his mind, has very low expected value. meet such investors last, if at all. [11]

doing breadth-first search weighted by expected value will save you from investors who never explicitly say no but merely drift away, because you'll drift away from them at the same rate. it protects you from investors who flake in much the same way that a distributed algorithm protects you from processors that fail. if some investor isn't returning your emails, or wants to have lots of meetings but isn't progressing toward making you an offer, you automatically focus less on them. but you have to be disciplined about assigning probabilities. you can't let how much you want an investor influence your estimate of how much they want you.

know where you stand.

how do you judge how well you're doing with an investor, when investors habitually seem more positive than they are? by looking at their actions rather than their words. every investor has some track they need to move along from the first conversation to wiring the money, and you should always know what that track consists of, where you are on it, and how fast you're moving forward.

never leave a meeting with an investor without asking what happens next. what more do they need in order to decide? do they need another meeting with you? to talk about what? and how soon? do they need to do something internally, like talk to their partners, or investigate some issue? how long do they expect it to take? don't be too pushy, but know where you stand. if investors are vague or resist answering such questions, assume the worst; investors who are seriously interested in you will usually be happy to talk about what has to happen between now and wiring the money, because they're already running through that in their heads. [12]

if you're experienced at negotiations, you already know how to ask such questions. [13] if you're not, there's a trick you can use in this situation. investors know you're inexperienced at raising money. inexperience there doesn't make you unattractive. being a noob at technology would, if you're starting a technology startup, but not being a noob at fundraising. larry and sergey were noobs at fundraising. so you can just confess that you're inexperienced at this and ask how their process works and where you are in it. [14]

get the first commitment.

the biggest factor in most investors' opinions of you is the opinion of other investors. once you start getting investors to commit, it becomes increasingly easy to get more to. but the other side of this coin is that it's often hard to get the first commitment.

getting the first substantial offer can be half the total difficulty of fundraising. what counts as a substantial offer depends on who it's from and how much it is. money from friends and family doesn't usually count, no matter how much. but if you get $50k from a well known vc firm or angel investor, that will usually be enough to set things rolling. [15]

close committed money.

it's not a deal till the money's in the bank. i often hear inexperienced founders say things like "we've raised $800,000," only to discover that zero of it is in the bank so far. remember the twin fears that torment investors? the fear of missing out that makes them jump early, and the fear of jumping onto a

when you first start fundraising, the expected value of an investor who won't "lead" is zero, so talk to such investors last if at all.

have multiple plans.

many investors will ask how much you're planning to raise. this question makes founders feel they should be planning to raise a specific amount. but in fact you shouldn't. it's a mistake to have fixed plans in an undertaking as unpredictable as fundraising.

so why do investors ask how much you plan to raise? for much the same reasons a salesperson in a store will ask "how much were you planning to spend?" if you walk in looking for a gift for a friend. you probably didn't have a precise amount in mind; you just want to find something good, and if it's inexpensive, so much the better. the salesperson asks you this not because you're supposed to have a plan to spend a specific amount, but so they can show you only things that cost the most you'll pay.

similarly, when investors ask how much you plan to raise, it's not because you're supposed to have a plan. it's to see whether you'd be a suitable recipient for the size of investment they like to make, and also to judge your ambition, reasonableness, and how far you are along with fundraising.

if you're a wizard at fundraising, you can say "we plan to raise a $7 million series a round, and we'll be accepting termsheets next tuesday." i've known a handful of founders who could pull that off without having vcs laugh in their faces. but if you're in the inexperienced but earnest majority, the solution is analogous to the solution i recommend for pitching your startup: do the right thing and then just tell investors what you're doing.

and the right strategy, in fundraising, is to have multiple plans depending on how much you can raise. ideally you should be able to tell investors something like: we can make it to profitability without raising any more money, but if we raise a few hundred thousand we can hire one or two smart friends, and if we raise a couple million, we can hire a whole engineering team, etc.

different plans match different investors. if you're talking to a vc firm that only does series a rounds (though there are few of those left), it would be a waste of time talking about any but your most expensive plan. whereas if you're talking to an angel who invests $20k at a time and you haven't raised any money yet, you probably want to focus on your least expensive plan.

if you're so fortunate as to have to think about the upper limit on what you should raise, a good rule of thumb is to multiply the number of people you want to hire times $15k times 18 months. in most startups, nearly all the costs are a function of the number of people, and $15k per month is the conventional total cost (including benefits and even office space) per person. $15k per month is high, so don't actually spend that much. but it's ok to use a high estimate when fundraising to add a margin for error. if you have additional expenses, like manufacturing, add in those at the end. assuming you have none and you think you might hire 20 people, the most you'd want to raise is 20 x $15k x 18 = $5.4 million.[18]

underestimate how much you want.

though you can focus on different plans when talking to different types of investors, you should on the whole err on the side of underestimating the amount you hope to raise.

for example, if you'd like to raise $500k, it's better to say initially that you're trying to raise $250k. then when you reach $150k you're more than half done. that sends two useful signals to investors: that you're doing well, and that they have to decide quickly because you're running out of room. whereas if

that matters. the real test is revenue. fundraising is just a means to that end. being proud of how well you did at fundraising is like being proud of your college grades.

not only is fundraising not the test that matters, valuation is not even the thing to optimize about fundraising. the number one thing you want from phase 2 fundraising is to get the money you need, so you can get back to focusing on the real test, the success of your company. number two is good investors. valuation is at best third.

the empirical evidence shows just how unimportant it is. dropbox and airbnb are the most successful companies we've funded so far, and they raised money after y combinator at premoney valuations of $4 million and $2.6 million respectively. prices are so much higher now that if you can raise money at all you'll probably raise it at higher valuations than dropbox and airbnb. so let that satisfy your competitiveness. you're doing better than dropbox and airbnb! at a test that doesn't matter.

when you start fundraising, your initial valuation (or valuation cap) will be set by the deal you make with the first investor who commits. you can increase the price for later investors, if you get a lot of interest, but by default the valuation you got from the first investor becomes your asking price.

so if you're raising money from multiple investors, as most companies do in phase 2, you have to be careful to avoid raising the first from an over-eager investor at a price you won't be able to sustain. you can of course lower your price if you need to (in which case you should give the same terms to investors who invested earlier at a higher price), but you may lose a bunch of leads in the process of realizing you need to do this.

what you can do if you have eager first investors is raise money from them on an uncapped convertible note with an mfn clause. this is essentially a way of saying that the valuation cap of the note will be determined by the next investors you raise money from.

it will be easier to raise money at a lower valuation. it shouldn't be, but it is. since phase 2 prices vary at most 10x and the big successes generate returns of at least 100x, investors should pick startups entirely based on their estimate of the probability that the company will be a big success and hardly at all on price. but although it's a mistake for investors to care about price, a significant number do. a startup that investors seem to like but won't invest in at a cap of $x will have an easier time at $x/2. [20]

yes/no before valuation.

some investors want to know what your valuation is before they even talk to you about investing. if your valuation has already been set by a prior investment at a specific valuation or cap, you can tell them that number. but if it isn't set because you haven't closed anyone yet, and they try to push you to name a price, resist doing so. if this would be the first investor you've closed, then this could be the tipping point of fundraising. that means closing this investor is the first priority, and you need to get the conversation onto that instead of being dragged sideways into a discussion of price.

fortunately there is a way to avoid naming a price in this situation. and it is not just a negotiating trick; it's how you (both) should be operating. tell them that valuation is not the most important thing to you and that you haven't thought much about it, that you are looking for investors you want to partner with and who want to partner with you, and that you should talk first about whether they want to invest at all. then if they decide they do want to invest, you can figure out a price. but first things first.

since valuation isn't that important and getting fundraising rolling is, we usually tell founders to give the first investor who commits as low a price as they need to. this is a safe technique so long as you combine it with the next one. [21]

you'll pick them. but lower-tier investors sometimes give offers with very short fuses, because they believe no one who had other options would choose them. a deadline of three working days is acceptable. you shouldn't need more than that if you've been talking to investors in parallel. but a deadline any shorter is a sign you're dealing with a sketchy investor. you can usually call their bluff, and you may need to. [22]

it might seem that instead of accepting offers greedily, your goal should be to get the best investors as partners. that is certainly a good goal, but in phase 2 "get the best investors" only rarely conflicts with "accept offers greedily," because the best investors don't usually take any longer to decide than the others. the only case where the two strategies give conflicting advice is when you have to forgo an offer from an acceptable investor to see if you'll get an offer from a better one. if you talk to investors in parallel and push back on exploding offers with excessively short deadlines, that will almost never happen. but if it does, "get the best investors" is in the average case bad advice. the best investors are also the most selective, because they get their pick of all the startups. they reject nearly everyone they talk to, which means in the average case it's a bad trade to exchange a definite offer from an acceptable investor for a potential offer from a better one.

(the situation is different in phase 1. you can't apply to all the incubators in parallel, because some offset their schedules to prevent this. in phase 1, "accept offers greedily" and "get the best investors" do conflict, so if you want to apply to multiple incubators, you should do it in such a way that the ones you want most decide first.)

sometimes when you're raising money from multiple investors, a series a will emerge out of those conversations, and these rules even cover what to do in that case. when an investor starts to talk to you about a series a, keep taking smaller investments till they actually give you a termsheet. there's no practical difficulty. if the smaller investments are on convertible notes, they'll just convert into the series a round. the series a investor won't like having all these other random investors as bedfellows, but if it bothers them so much they should get on with giving you a termsheet. till they do, you don't know for sure they will, and the greedy algorithm tells you what to do. [23]

don't sell more than 25% in phase 2.

if you do well, you will probably raise a series a round eventually. i say probably because things are changing with series a rounds. startups may start to skip them. but only one company we've funded has so far, so tentatively assume the path to huge passes through an a round. [24]

which means you should avoid doing things in earlier rounds that will mess up raising an a round. for example, if you've sold more than about 40% of your company total, it starts to get harder to raise an a round, because vcs worry there will not be enough stock left to keep the founders motivated.

our rule of thumb is not to sell more than 25% in phase 2, on top of whatever you sold in phase 1, which should be less than 15%. if you're raising money on uncapped notes, you'll have to guess what the eventual equity round valuation might be. guess conservatively.

(since the goal of this rule is to avoid messing up the series a, there's obviously an exception if you end up raising a series a in phase 2, as a handful of startups do.)

have one person handle fundraising.

if you have multiple founders, pick one to handle fundraising so the other(s) can keep working on the company. and since the danger of fundraising is not the time taken up by the actual meetings but that

through a straw, you can tell when you get to the end of the liquid because you start to get a lot of air in the straw. when your fundraising options run out, they usually run out in the same way. don't keep sucking on the straw if you're just getting air. it's not going to get better.

don't get addicted to fundraising.

fundraising is a chore for most founders, but some find it more interesting than working on their start-up. the work at an early stage startup often consists of unglamorous schleps. whereas fundraising, when it's going well, can be quite the opposite. instead of sitting in your grubby apartment listening to users complain about bugs in your software, you're being offered millions of dollars by famous investors over lunch at a nice restaurant. [26]

the danger of fundraising is particularly acute for people who are good at it. it's always fun to work on something you're good at. if you're one of these people, beware. fundraising is not what will make your company successful. listening to users complain about bugs in your software is what will make you successful. and the big danger of getting addicted to fundraising is not merely that you'll spend too long on it or raise too much money. it's that you'll start to think of yourself as being already successful, and lose your taste for the schleps you need to undertake to actually be successful. startups can be destroyed by this.

when i see a startup with young founders that is fabulously successful at fundraising, i mentally decrease my estimate of the probability that they'll succeed. the press may be writing about them as if they'd been anointed as the next google, but i'm thinking "this is going to end badly."

don't raise too much.

though only a handful of startups have to worry about this, it is possible to raise too much. the dangers of raising too much are subtle but insidious. one is that it will set impossibly high expectations. if you raise an excessive amount of money, it will be at a high valuation, and the danger of raising money at too high a valuation is that you won't be able to increase it sufficiently the next time you raise money.

a company's valuation is expected to rise each time it raises money. if not it's a sign of a company in trouble, which makes you unattractive to investors. so if you raise money in phase 2 at a post-money valuation of $30 million, the pre-money valuation of your next round, if you want to raise one, is going to have to be at least $50 million. and you have to be doing really, really well to raise money at $50 million.

it's very dangerous to let the competitiveness of your current round set the performance threshold you have to meet to raise your next one, because the two are only loosely coupled.

but the money itself may be more dangerous than the valuation. the more you raise, the more you spend, and spending a lot of money can be disastrous for an early stage startup. spending a lot makes it harder to become profitable, and perhaps even worse, it makes you more rigid, because the main way to spend money is people, and the more people you have, the harder it is to change directions. so if you do raise a huge amount of money, don't spend it. (you will find that advice almost impossible to follow, so hot will be the money burning a hole in your pocket, but i feel obliged at least to try.)

be nice.

startups raising money occasionally alienate investors by seeming arrogant. sometimes because they are arrogant, and sometimes because they're noobs clumsily attempting to mimic the toughness they've observed in experienced founders.

justify them. a lot of vcs will encourage you to hire aggressively. vcs generally tell you to spend too much, partly because as money people they err on the side of solving problems by spending money, and partly because they want you to sell them more of your company in subsequent rounds. don't listen to them.

don't make things complicated.

i realize it may seem odd to sum up this huge treatise by saying that my overall advice is not to make fundraising too complicated, but if you go back and look at this list you'll see it's basically a simple recipe with a lot of implications and edge cases. avoid investors till you decide to raise money, and then when you do, talk to them all in parallel, prioritized by expected value, and accept offers greedily. that's fundraising in one sentence. don't introduce complicated optimizations, and don't let investors introduce complications either.

fundraising is not what will make you successful. it's just a means to an end. your primary goal should be to get it over with and get back to what will make you successful — making things and talking to users — and the path i've described will for most startups be the surest way to that destination.

be good, take care of yourselves, and *don't leave the path*.

---

notes

1 the worst explosions happen when unpromising-seeming startups encounter mediocre investors. good investors don't lead startups on; their reputations are too valuable. and startups that seem promising can usually get enough money from good investors that they don't have to talk to mediocre ones. it is the unpromising-seeming startups that have to resort to raising money from mediocre investors. and it's particularly damaging when these investors flake, because unpromising-seeming startups are usually more desperate for money.

(not all unpromising-seeming startups do badly. some are merely ugly ducklings in the sense that they violate current startup fashions.)

2 one yc founder told me:

> i think in general we've done ok at fundraising, but i managed to screw up twice at the exact same thing — trying to focus on building the company and fundraising at the same time.

3 there is one subtle danger you have to watch out for here, which i warn about later: beware of getting too high a valuation from an eager investor, lest that set an impossibly high target when raising additional money.

4 if they really need a meeting, then they're not ready to invest, regardless of what they say. they're still deciding, which means you're being asked to come in and convince them. which is fundraising.

5 associates at vc firms regularly cold email startups. naive founders think "wow, a vc is interested in us!" but an associate is not a vc. they have no decision-making power. and while they may introduce startups they like to partners at their firm, the partners discriminate against deals that come to them this way. i don't know of a single vc investment that began with an associate cold-emailing a startup. if you want to approach a specific firm, get an intro to a partner from someone they respect.

it's ok to talk to an associate if you get an intro to a vc firm or they see you at a demo day and they begin by having an associate vet you. that's not a promising lead and should therefore get low priority, but it's not as completely worthless as a cold email.

16 sometimes a competitor will deliberately threaten you with a lawsuit just as you start fundraising, because they know you'll have to disclose the threat to potential investors and they hope this will make it harder for you to raise money. if this happens it will probably frighten you more than investors. experienced investors know about this trick, and know the actual lawsuits rarely happen. so if you're attacked in this way, be forthright with investors. they'll be more alarmed if you seem evasive than if you tell them everything.

17 a related trick is to claim that they'll only invest contingently on other investors doing so because otherwise you'd be "undercapitalized." this is almost always bullshit. they can't estimate your minimum capital needs that precisely.

18 you won't hire all those 20 people at once, and you'll probably have some revenues before 18 months are out. but those too are acceptable or at least accepted additions to the margin for error.

19 type a fundraising is so much better that it might even be worth doing something different if it gets you there sooner. one yc founder told me that if he were a first-time founder again he'd "leave ideas that are up-front capital intensive to founders with established reputations."

20 i don't know whether this happens because they're innumerate, or because they believe they have zero ability to predict startup outcomes (in which case this behavior at least wouldn't be irrational). in either case the implications are similar.

21 if you're a yc startup and you have an investor who for some reason insists that you decide the price, any yc partner can estimate a market price for you.

22 you should respond in kind when investors behave upstandingly too. when an investor makes you a clean offer with no deadline, you have a moral obligation to respond promptly.

23 tell the investors talking to you about an a round about the smaller investments you raise as you raise them. you owe them such updates on your cap table, and this is also a good way to pressure them to act. they won't like you raising other money and may pressure you to stop, but they can't legitimately ask you to commit to them till they also commit to you. if they want you to stop raising money, the way to do it is to give you a series a termsheet with a no-shop clause.

you can relent a little if the potential series a investor has a great reputation and they're clearly working fast to get you a termsheet, particularly if a third party like yc is involved to ensure there are no misunderstandings. but be careful.

24 the company is weebly, which made it to profitability on a seed investment of $650k. they did try to raise a series a in the fall of 2008 but (no doubt partly because it was the fall of 2008) the terms they were offered were so bad that they decided to skip raising an a round.

25 another advantage of having one founder take fundraising meetings is that you never have to negotiate in real time, which is something inexperienced founders should avoid. one yc founder told me:

> investors are professional negotiators and can negotiate on the spot very easily. if only one founder is in the room, you can say "i need to circle back with my co-founder" before making any commitments. i used to do this all the time.

26 you'll be lucky if fundraising feels pleasant enough to become addictive. more often you have to worry about the other extreme — becoming demoralized when investors reject you. as one (very successful) yc founder wrote after reading a draft of this:

> it's hard to mentally deal with the sheer scale of rejection in fundraising and if you are not in the right mindset you will fail. users may love you but these supposedly smart investors may not understand you at all. at this point for me, rejection still rankles but i've come to accept that investors are just not super thoughtful for the most part and you need to play the game according to certain somewhat depressing rules (many of which you are listing) in order to win.

12

donate unrestricted

march 2021
paulgraham.com/donate.html